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Department of CSE

B.Tech. CSE Trimester -5

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Computer Networks and Network Programming

25CS2208E

LAB RECORD

2026

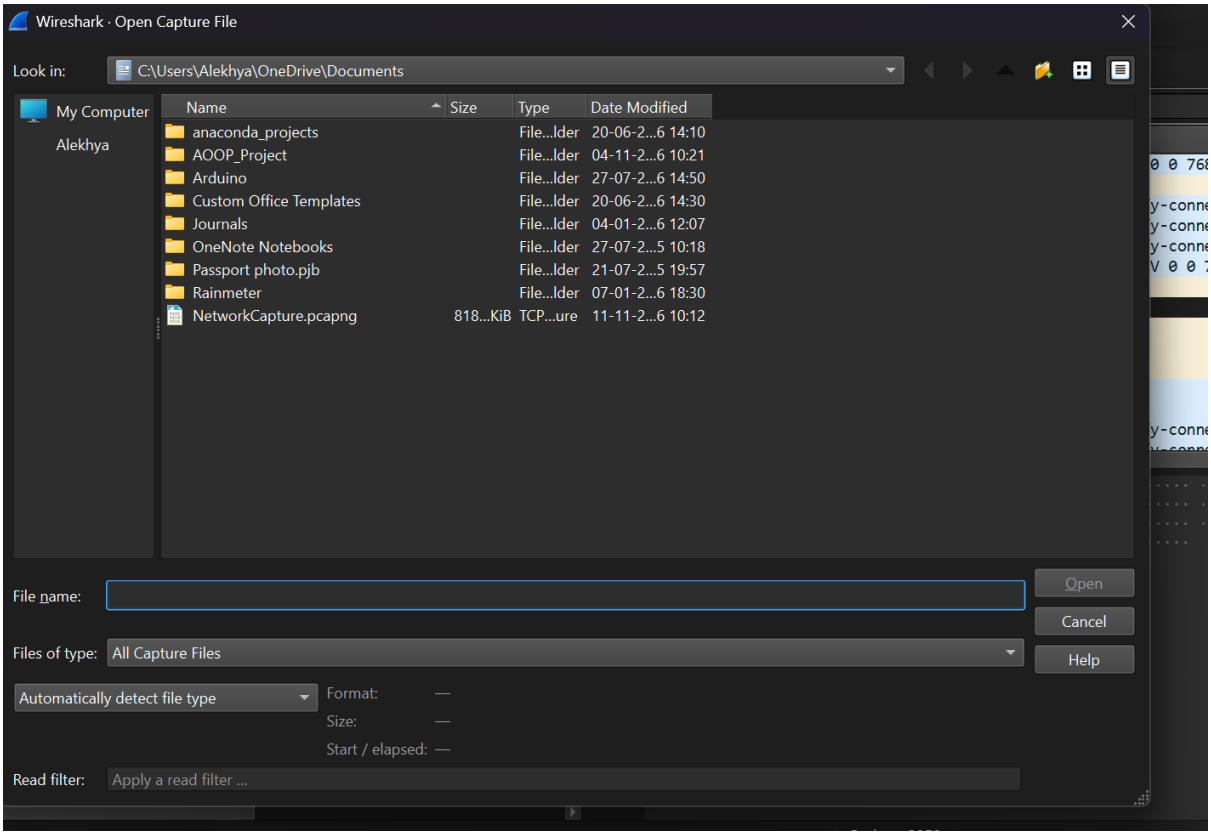
Experiment -1

AIM: To analyze a PCAP (Packet Capture) file using Wireshark, identify the total number of captured packets, determine the number of TCP, UDP, and ICMP packets using display filters, and identify the top five protocols using the Protocol Hierarchy statistics.

Steps:

Step 1:Open the PCAP file

1. Open Wireshark.
2. Click on File → Open.
3. Select the required PCAP file.
4. Click Open to load the captured packets.
5. Observe the packets displayed in the Wireshark window.

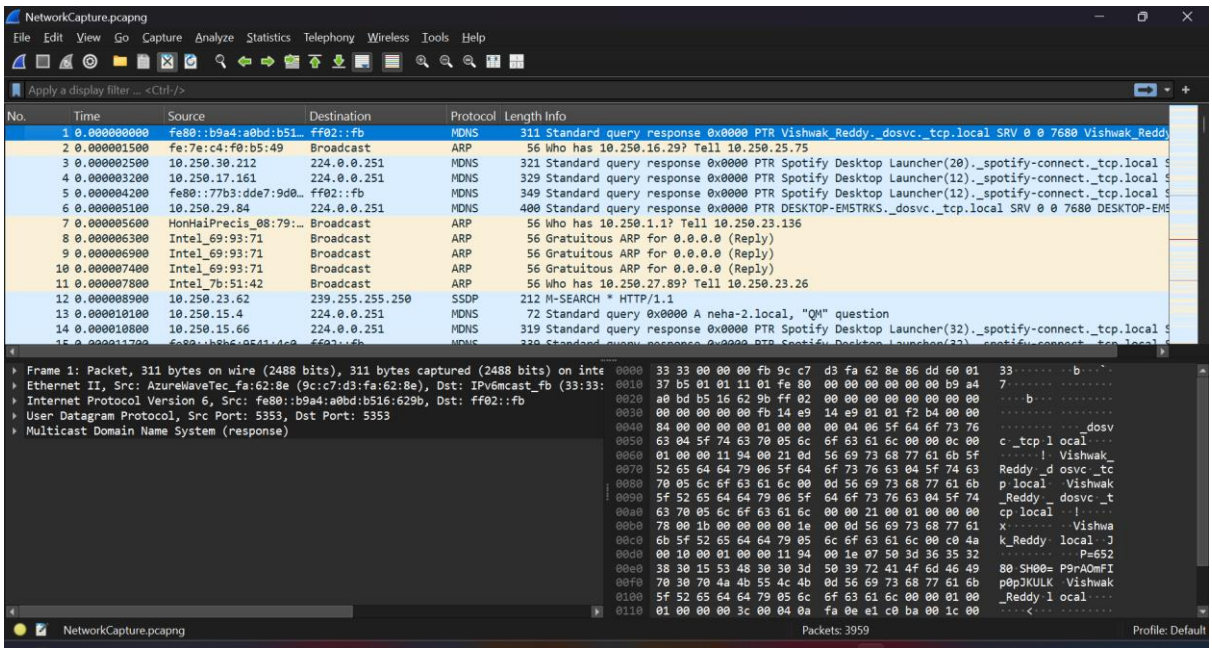


Step 2: Identify total number of packets

1. Open the PCAP file in Wireshark.
2. Look at the bottom of the Wireshark window.
3. Find the packet count displayed in the status bar.

4. Record the total number of packets.

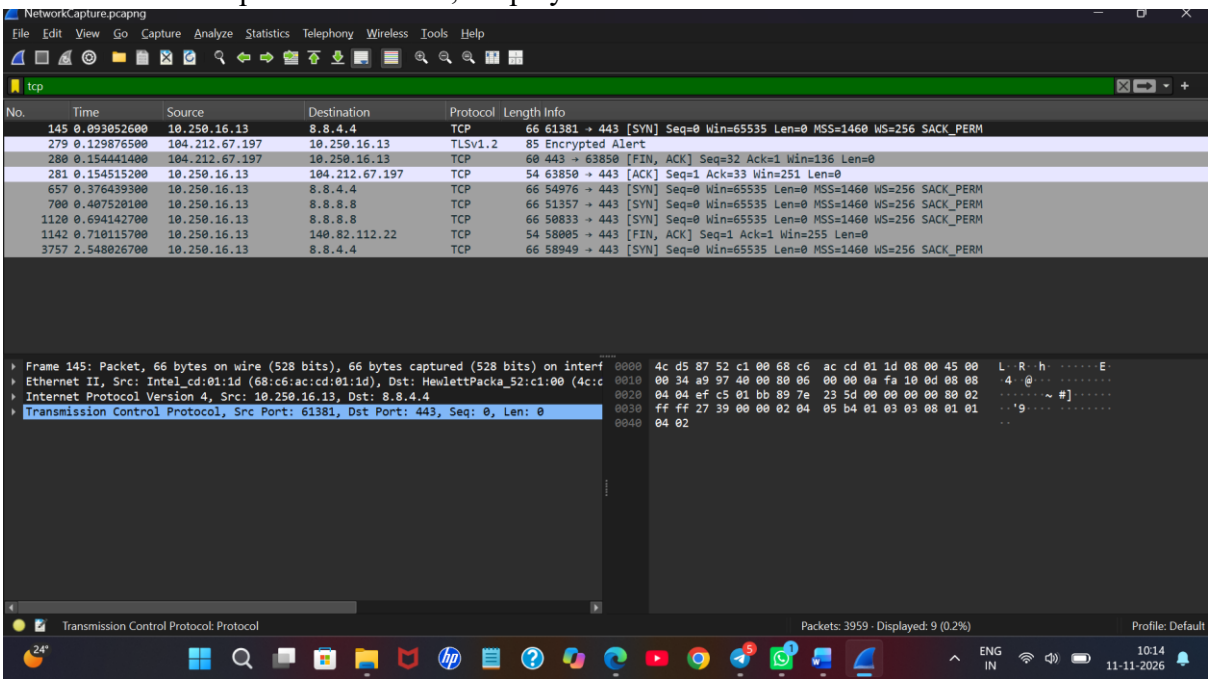
Result: Total packets captured = 3959



Step 3: Identify TCP packets

1. Click the display filter bar at the top.
2. Type the filter:tcp
3. Press Enter.
4. Wireshark displays only TCP packets.
5. Check the packet count at the bottom.
6. Record the number.

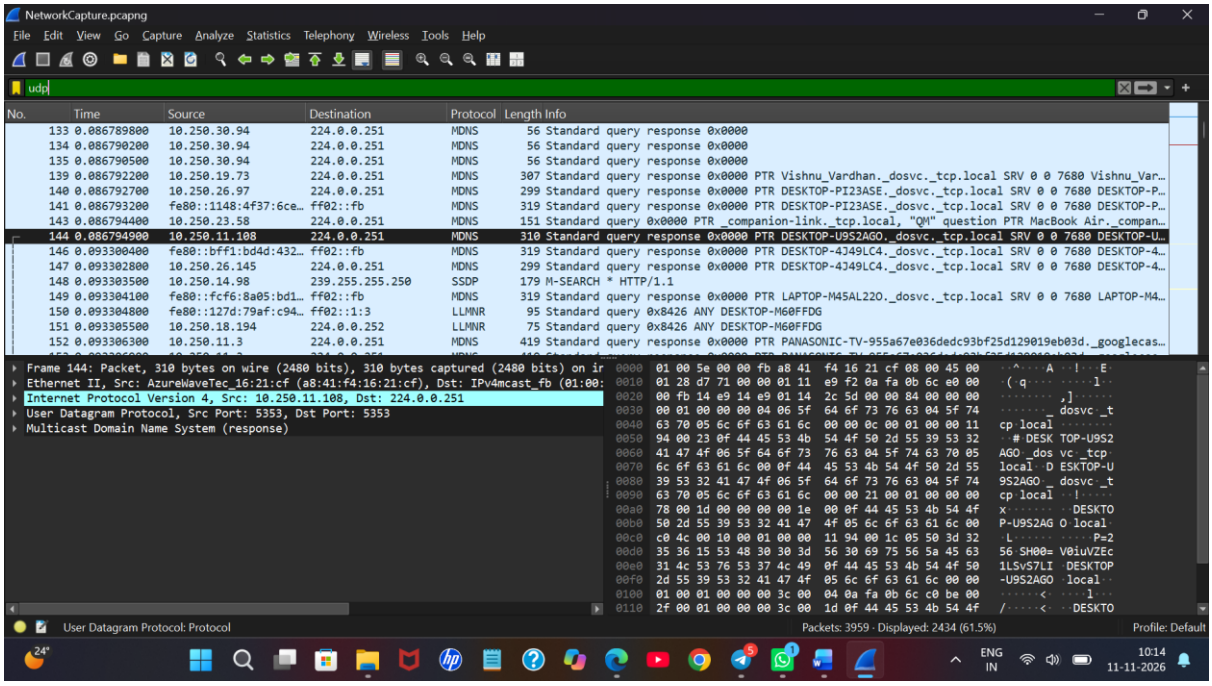
Result: Total TCP packets = 34518, Displayed = 3959



Step 4: Identify UDP packets

1. Click the display filter bar.
2. Remove the previous filter.
3. Type:udp
4. Press Enter.
5. Only UDP packets will be displayed.
6. Check the packet count at the bottom.
7. Record the number.

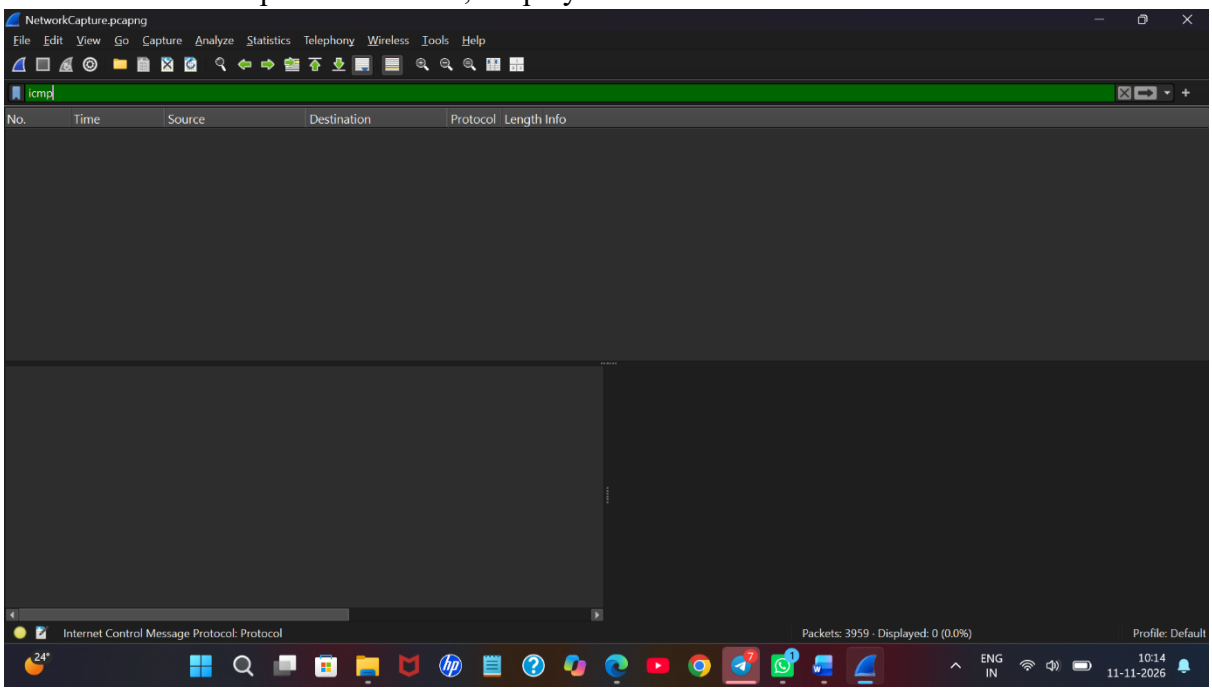
   Result: Total UDP packets = 3959, Displayed = 2434



Step 5: Identify ICMP packets

1. Click the display filter bar.
2. Remove the previous filter.
3. Type:icmp
4. Press Enter.
5. Only ICMP packets will be displayed.
6. Check the packet count at the bottom.
7. Record the number.

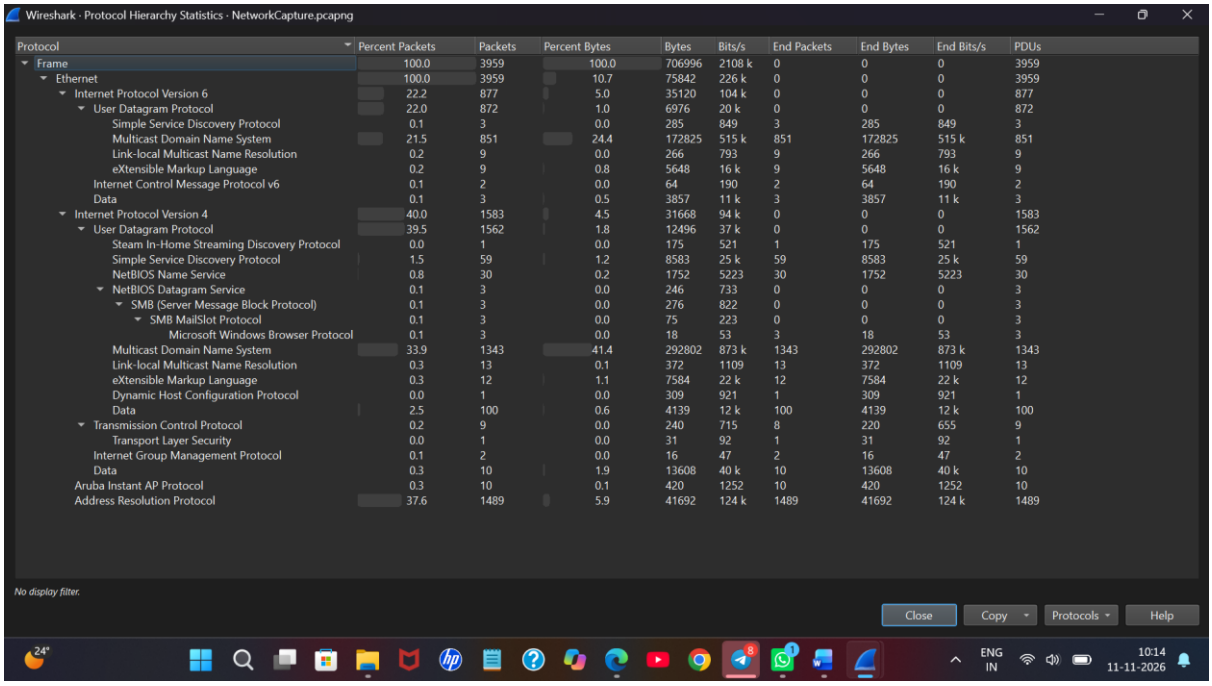
Result: Total ICMP packets = 3959 , Displayed = 0



Step 6: Identify the top five protocols

1. Click on Statistics in the menu bar.
2. Select Protocol Hierarchy.
3. A new window will appear.
4. Observe the list of protocols.
5. Identify the top five protocols based on packet count or percentage.
6. Record their names and corresponding packet counts.
7. Take a screenshot for your record.

Protocol	Packet Count
Multicast DNS (mDNS)	2312
ICMPv6	523
Simple Service Discovery Protocol (SSDP)	130
Link-Local Multicast Name Resolution (LLMNR)	126
NetBIOS Name Service	54



Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	3959	100.0	706996	2108 k	0	0	0	3959
Ethernet	100.0	3959	10.7	75842	226 k	0	0	0	3959
Internet Protocol Version 6	22.2	877	5.0	35120	104 k	0	0	0	877
User Datagram Protocol	22.0	872	1.0	6976	20 k	0	0	0	872
Simple Service Discovery Protocol	0.1	3	0.0	285	849	3	285	849	3
Multicast Domain Name System	21.5	851	24.4	172825	515 k	851	172825	515 k	851
Link-local Multicast Name Resolution	0.2	9	0.0	266	793	9	266	793	9
eXtensible Markup Language	0.2	9	0.8	5648	16 k	9	5648	16 k	9
Internet Control Message Protocol v6	0.1	2	0.0	64	190	2	64	190	2
Data	0.1	3	0.5	3857	11 k	3	3857	11 k	3
Internet Protocol Version 4	40.0	1583	4.5	31668	94 k	0	0	0	1583
User Datagram Protocol	39.5	1562	1.8	12496	37 k	0	0	0	1562
Steam In-Home Streaming Discovery Protocol	0.0	1	0.0	175	521	1	175	521	1
Simple Service Discovery Protocol	1.5	59	1.2	8583	25 k	59	8583	25 k	59
NetBIOS Name Service	0.8	30	0.2	1752	5223	30	1752	5223	30
NetBIOS Datagram Service	0.1	3	0.0	246	733	0	0	0	3
SMB (Server Message Block Protocol)	0.1	3	0.0	276	822	0	0	0	3
SMB MailSlot Protocol	0.1	3	0.0	75	223	0	0	0	3
Microsoft Windows Browser Protocol	0.1	3	0.0	18	53	3	18	53	3
Multicast Domain Name System	33.9	1343	41.4	292802	873 k	1343	292802	873 k	1343
Link-local Multicast Name Resolution	0.3	13	0.1	372	1109	13	372	1109	13
eXtensible Markup Language	0.3	12	1.1	7584	22 k	12	7584	22 k	12
Dynamic Host Configuration Protocol	0.0	1	0.0	309	921	1	309	921	1
Data	2.5	100	0.6	4139	12 k	100	4139	12 k	100
Transmission Control Protocol	0.2	9	0.0	240	715	8	220	655	9
Transport Layer Security	0.0	1	0.0	31	92	1	31	92	1
Internet Group Management Protocol	0.1	2	0.0	16	47	2	16	47	2
Data	0.3	10	1.9	13608	40 k	10	13608	40 k	10
Aruba Instant AP Protocol	0.3	10	0.1	420	1252	10	420	1252	10
Address Resolution Protocol	37.6	1489	5.9	41692	124 k	1489	41692	124 k	1489